

Q-Orbit — Phase 1 Device-Imperfection Mapping (DELIVERABLE 5)

Document ID: QO-P1-D5 | **Version:** 1.0 | **Date:** 2026-08-27 **Inputs:** Agent C source-imperfection brief (S1–S8); Agent D detector/receiver brief (12 effects); Agent E characterization-to-proof bridge (pipeline, security budget, triage); Canonical Facts Record QO-CFR-001 v1.0 (locked ground truth); Phase 1 Canonical Audit. **Policy:** fail-closed. No numerical value is invented in this document; the only numerics quoted are locked values from the CFR, cited as CFR N-xx. Status labels are drawn verbatim from the controlled set: MAPPED-IN-CURRENT-FIXTURE / PARTIAL-SCALAR-STRESS-ONLY / PROOF-PROFILE-CANDIDATE / UNMAPPED-PROOF-REQUIRED / UNMAPPED-MODEL-REQUIRED / UNMAPPED-CHARACTERIZATION-REQUIRED / UNMAPPED-SECURITY-BUDGET / BLOCKING.

1. Purpose and method

Purpose. Produce the single consolidated mapping between the 15 mandated device-imperfection effects and (a) the frozen V0.16-TA1 scalar software fixture, (b) the security-proof literature that could treat each effect, and (c) the characterization evidence each treatment would require. This document is the merge point of the C (source), D (detector/receiver), and E (characterization/statistics) briefs and is the input to the V0.17-TA1 architecture specification (Deliverable 6).

Context (locked). Q-Orbit is a theoretical satellite-QKD package: efficient (biased-basis) BB84, weak-coherent-pulse downlink, 1 signal + 2 decoy intensities (one vacuum); finite-key margin equation after the Sidhu et al. (npj Quantum Information 8, 18, 2022 — VERIFIED) fixture with the Lim et al. (PRA 89, 022307, 2014 — VERIFIED) structure: $M = s_X, \theta + s_X, 1 \cdot [1 - h_2(\phi_X)] - \lambda_{EC} - 6 \cdot \log_2(21/\epsilon_s) - \log_2(2/\epsilon_c)$. The software contract carries 8 scalars (TH-PAR-001 additional loss ... TH-PAR-008 intrinsic QBER). There is **zero physical characterization** (CFR B-01/B-02: `physical_characterization: NOT-EXECUTED`; `hardware_in_loop: BLOCKED`). The frozen fixture's own audit records a 16-row imperfection-to-proof mapping with 7 rows unmapped (CFR via audit AUD-016-010; the mapping CSV itself is REQ-05, hash-recorded but not in the bundle).

Method.

1. Row construction: the 15 mandated effects are partitioned as source-side (effects 1–5: Agent C S1–S7), detector/receiver-side (effects 6–13: Agent D §1–§11), and cross-cutting (effects 14–15: Agent C S8 + Agent E §1.3–§1.4). Agent C's S3 (independent intensity fluctuations) is not among the 15 mandated rows; it is carried as a note under effect 3 because it shares the TH-PAR-004/005 scalars and the same modulator hardware.
2. Classification rule (shared C/D guiding question): an effect is *engineering count-model only* if it changes observed rates/QBER without changing (i) the adversary's information, (ii) the validity of the source/detection model assumed by the proof (ideal qubit encoding, photon-number channel structure `_n`, IID intensities, perfect phase randomization, no side channels, squashing, photon-number-independent efficiency, no memory), or (iii) the finite-key statistical statements. Otherwise it is **proof-modifying**.
3. Status assignment: each row carries the per-agent label(s); where C/D/E labels disagree, both are shown and the row is resolved to the **stricter (fail-closed)** label. No effect receives MAPPED-IN-CURRENT-FIXTURE — inspection of the frozen fixture (audit §C, EV-4) confirms no scalar fully represents any of the 15 effects.

2. MASTER MATRIX — the 15 mandated effects

#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathematical parameter	Required characterization evidence	Confidence treatment	Current status
1	Incomplete phase randomization	None — perfect PR assumed silently; no fixture scalar can express it	Proof-modifying, architectural: without (near-)uniform phase randomization the photon-number decomposition ρ_n does not exist, so $s_{X,0}$, $s_{X,1}$, ρ_X — every margin term except ρ_{EC} — is unfounded; non-random phases additionally enable phase-information attacks (Lo & Preskill, QIC 7, 431, 2007 — secondary-verified)	Nahar, Upadhyaya, Lütkenhaus, PR Applied 20, 064031 (2023) — VERIFIED (generalized decoy with characterized phase PDF); Currás-Lorenzo et al., QST 9, 015025 (2023) — VERIFIED; faulty active PR: Sixto et al., EPJ Quantum Technol. 10, 53 (2023) — VERIFIED	Characterized global-phase distribution / deviation-from-uniform bound within the proof's admissible class	Inter-pulse interferometric visibility; measured phase PDF; inter-pulse coherence time; verification the measured PDF lies inside the admissible class	Phase-PDF bound with CI at stated 1- σ ; fail-closed if unmeasured	UNFROZEN PROPOSED RECOMMENDED (frozen fixture as-is) PROPOSED CALIBRATION AVAILABLE GATED UNFROZEN CHALLENGE RECOMMENDED (no previous measurement exists) RESOLVED (strictly C's label) UNFROZEN PROPOSED RECOMMENDED

Table 1 – continued

#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
2	Pulse-to- pulse correla- tions (encoding memory)	None — fixture assumes IID pulses	Proof- modifying: breaks conditional independen- ce assumed by Hoeffding/ Serfling finite-key statistics, the decoy conditional- probability structure, and the random- sampling phase-error bound; gives Eve cross-round joint infor- mation	Nagamatsu et al., PRA 93, 042325 (2016) — VERIFIED via citation records; Mizutani et al., npj QI 5, 8 (2019) —VERI- FIED; Pereira et al., Sci. Adv. 6, eaaz4487 (2020) — VERIFIED via citation records; strongest: Pereira et al., QST 10, 015001 (2025) — VERIFIED (un- bounded correla- tions); Marwah & Dupuis, arXiv:2402.12346 (2024) — PREPRINT only, non-load- bearing; statistical layer: Azuma (Tohoku Math. J. 19, 357, 1967) / Kato	Correlation length and correlation- strength bound (joint over encoding DOF)	Conditional state to- mography given predecessor settings (pattern- dependence maps); autocorre- lation of emitted states vs lag; drift spectra	Martingale (Azuma/ Kato) bounds with bounded increments; CI on correlation- strength bound	UNI PRO REC + UNI CH REC (C' s no d ment

Table 1 – continued

#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
3	Intensity correla- tions (pulse-to- pulse)	None – TH-PAR- 004/005 are exact scalars; correlation struc- turally inexpress- ible	Proof- modifying: correlated intensities break the IID structure making $p_{\{k n\}}$ well- defined per pulse; correlation pattern is setting- dependent, hence setting- revealing to Eve. Experimen- tally demon- strated in deployed decoy systems (Yoshino et al., npj QI 4, 8, 2018 –VERI- FIED; Trefilov et al., arXiv:2411.00709, 2024 – PREPRINT only)	Zapatero, Navarrete, Curty, Quantum 5, 602 (2021) – VERIFIED (bounded nearest- neighbour); Sixto, Zapatero, Curty, PR Applied 18, 044069 (2022) – VERIFIED (correlated fluctua- tions); arbitrarily long via Pereira et al., QST 10, 015001 (2025)	Correlation length $_;$ bound on conditional- intensity deviation $p(_i $ settings of $i-1, \dots, i-$)	Measured conditional intensity distribu- tions vs predecessor settings; measure- ment; counter- measure validation if patterning mitigation is claimed	CI on $_$ and on the conditional- deviation bound	PRO PRO CAL gated UNI CH REC (C' s Note TH-I 004/ touch <i>adjae</i> effec depe inter fluct tions S3) a exac scala PAE SCA STR ONI there contr noth this corre row

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
4	State- preparation (encoding) flaws	Absorbed into scalar QBER/ misalign- ment contribu- tion (TH-PAR- 008 intrinsic QBER); only the basis- independent component is repre- sentable	Proof- modifying: basis- dependent flaws invalidate the GLLP Δ -term- free $_X$ bound; Eve’ s in- formation becomes basis- asymmetric while the $_X$ bound still assumes symmetry. GLLP (QIC 4, 325, 2004 –VERI- FIED) covers only basis- independent flaws; real flaws are generically basis- dependent. Measured in deployed systems: Xu et al., PRA 92, 032305 (2015) – VERIFIED (evidence about that device, not Q-Orbit’ s)	Loss- tolerant protocol: Tamaki et al., PRA 90, 052314 (2014) – VERI- FIED; finite-key incl. intensity fluctua- tions: Mizutani et al., NJP 17, 093011 (2015) – VERI- FIED; random- sampling finite-key variant: Currás- Lorenzo et al., PRA 104, 012406 (2021) – secondary- verified; modified BB84: Pereira et al., PRR 5, 023065 (2023) – VERI- FIED; unified framework: Currás- Lorenzo et al., Optica Quantum 3, 525 (2025) –	Bloch- sphere deviation $_spf$ / pairwise state overlaps; or measured quantum- coin imbalance Δ ; or loss- tolerant qubit-flaw parameters (three-state structure)	Tomographic or reference- state measure- ment of the four emitted density operators (minimum: pairwise overlaps + $_spf$ bound); evidence of basis- (in)dependence; stability over operating conditions	CI on $_spf$ enters the robust parameter set S_robust (Tan- Nahar, PRX Quantum 7, 020342, 2026 – VERI- FIED)	UNI PRO REQ (basis depe part) PRO PRO CAL avail the b indep comp folde TH-I 008 i PAR SCA STR ONI Reso stric UNM PRO REQ

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr
5	Source leakage / distinguishability (passive side channels + Trojan- horse active leakage)	None —no fixture scalar; leakage is outside the count/ QBER model entirely	Proof- modifying: setting- dependent side- channel modes (spectrum, timing, spatial mode, chirp) leak basis/bit/ intensity choices with zero QBER signature (generic in modulator transmit- ters: Gnanapan- dithan et al., PRL 134, 130802, 2025 — secondary- verified); active Trojan- horse injection reads modulator state via back- reflection (Gisin et al., PRA 73, 022320, 2006 — VERIFIED via citation records)	Passive modes: unified framework (Currás- Lorenzo et al., Optica Quantum 3, 525, 2025 — VERI- FIED); MDI passive side channels: Bourassa et al., PRA 106, 062618 (2022) — secondary- verified. Trojan horse (all conditional on measured isolation): Lucamarini et al., PRX 5, 031030 (2015) — VERI- FIED; Tamaki et al., NJP 18, 065008 (2016) — VERI- FIED; Wang et al., NJP 20, 083027 (2018) — VERI- FIED; Navarrete	Passive: mode- overlap / distinguishability bound per setting (4 states $\times$ 3 intensities). Active: isolation bound (dB) over all input ports; back- reflected mean photon number __out; aging worst-case	Spectral/ temporal/ spatial mode mea- surements conditioned on all 12 setting combina- tions; source isolation measure- ment; back- reflection coefficient; modulator response to injected light; worst-case bounds under aging	Distinguishability budget + CI; isolation CI enters S_robust; abort if unbounded	Cyber- UN- SEC- BUL- + UN- CHA- REC- (pass- and BLO- ING- unco- tiona- claim- while isola- unm- sure- (acti- Reso- stric- BLO- ING- (acti- leaka- comp- gove-

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
6	Dead time / recovery	None — detector- efficiency multiplier (TH-PAR- 002) is rate- independent; no time- stepped detector state	Engineering count- model only IF dead time/ recovery is identical across detectors, constant, non- adversarial —but escalates to proof- modifying because (i) efficiency becomes rate- dependent and dy- namically basis- dependent across the pass loss sweep (dy- namically generated mismatch, cf. row 10); (ii) adver- sorially exploitable: dead-time attack (Weier et al., NJP 13, 073024, 2011 — VERI- FIED); avalanche- transition- region	Engineering: explicit paralyz- able/ non- paralyzable per- detector count model; receiver finite-state- machine model with measured transition probabili- ties. Security: bounded time- dependent mismatch proofs (Fung et al., QIC 9, 131, 2009 — VERI- FIED; Zhang et al., PRR 3, 013076, 2021 — VERI- FIED; Bochkov & Trushechkin, PRA 99, 032308, 2019 — VERI- FIED; Trushechkin, Quantum 6, 771, 2022 —	Per- detector dead-time distribu- tion; recovery curve (Δt); rate- dependent yield function; latching thresholds	Per- detector dead-time/ recovery curves (double-/ triple-pulse efficiency vs separation); paralyzable vs non- paralyzable identifica- tion from count-rate curves; bright- pulse response (adversarial resilience); basis- independence cross-check	CI per curve over rate/tem- perature envelope; mechanism confidence high; proof treatment for adversarial case partial (preprint- level)	D la UNI MO REC (engi ing) UNI SEC BUI (adve dead attac Reso stric UNM SEC BUI TH-I 002 PA SCA STR ONI (unif low- equil only)

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
7	Saturation	None — fixture count model is linear in ; rollover struc- turally inexpress- ible	Engineering- only when saturation is identical across detectors, reached only by honest sig- nal+background and monitored. Proof- modifying whenever detectors saturate differently (dynamic rate- dependent mismatch) or saturation is approached by injected light —the onset of the detector- control class (Makarov, NJP 11, 065003, 2009; Sauge et al., Opt. Express 19, 23590, 2011; Lydersen et al., NJP 13, 113042, 2011 —all	Engineering: nonlinear response model R_obs(_in) per detector. Security: certified maximum input flux as device assumption + charac- terized power limiter (security- boundary analysis, arXiv:2303.12355 —VERI- FIED); residual sub-limit nonlinear- ity enters as bounded efficiency mismatch	Nonlinear response function per detector; certified max-flux bound; limiter transfer function	Per- detector full input- output curves from single- photon level through saturation and overload recovery; TDC/ readout throughput limits	CI over the full flux envelope; abort on any excursion beyond certified range	D lab UNI MO REC (engi ing); BLO ING the l regin entry of th dete contr class Reso stric BLO ING TH-I 002 none sent beyo linea regin

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
8	Detector timing jitter	Window- average of (t) can inform TH- PAR-002; the time- resolved structure is absent	Partially proof- modifying: jitter converts arrival time —an Eve- modulable DOF —into detection probability; inter- detector/ basis jitter asymmetry yields time- dependent efficiency mismatch, the exact enabler of the time-shift attack (Qi et al., QIC 7, 73, 2007 —VERI- FIED; demon- strated: Zhao et al., PRA 78, 042333, 2008 — VERI- FIED); timing side channel: Lamas- Linares & Kurtsiefer, Opt. Express 15, 9388, 2007 — VERIFIED.	Time- resolved mismatch- bounded proofs (Fung 2009; Zhang PRR 2021; Bochkov- Trushechkin PRA 2019); architec- tural elimination via single detector + fast basis switch; caution — random- detector- efficiency counter- measure demon- strated breakable (Huang et al., IEEE JQE 52, 8000411, 2016 — VERI- FIED)	Per- detector, per-basis (t) across the acceptance window; worst-case over Eve- controlled arrival times	Per- detector timing- jitter histograms (instrument response functions); efficiency- vs-arrival- delay maps; stability vs tempera- ture, count rate, history	CI on (t) maps over the repetition envelope (§E- matrix)	D la UNI CH REC (proc mach exist input meas TH-I 002 t PA SCA STR ONI (win avera only)

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
9	History- dependent afterpuls- ing	TH-PAR- 007 afterpulse scalar = equilib- rium IID marginal only (first-order stationary trap occu- pancy); contributes to average QBER like extra dark counts	History/ rate/basis- dependent part is proof- modifying; afterpuls- ing is P(click in slot i detections in slots $<$ i) with power-law/ multi- timescale memory (Ziarkash et al., Sci. Rep. 8, 5076, 2018; Itzler et al., J. Mod. Opt. 59, 1472, 2012; Horoshko et al., J. Mod. Opt. 64, 191, 2017 –all VERI- FIED); joint click statistics non-IID $\rightarrow$ Serfling random sampling not directly valid for this component; same- detector correlated clicks	Engineering: trap-kernel conditional click model (sum-of- exponentials or power-law release; estimation: Humer et al., JLT 33, 3098, 2015 –VERI- FIED). Security: correlated- noise process with martingale finite-key bounds (Azuma 1967; Kato arXiv:2002.04357); or gated hold-off with per-gate conditional- probability characteri- zation. A theorem that worst-case IID over- approximation is pessimistic for this correlation does not exist in verified	Conditional click- probability kernel P(click history); multi- timescale release law; p_ap(rate) function	Lag- resolved conditional click proba- bilities per detector (full memory kernel); p_ap vs count rate; tempera- ture/bias depen- dence; afterpulse response to injected bright pulses	Martingale (Azuma/ Kato) bounds; models peer- reviewed (high confidence); proof-level treatment of correlated back- ground: open (§6)	D lab PAR- SCA STR ONL the s histo depe rema UNI PRO REC Reso UNM PRO REQ (stric with scala corre denc expli recon PAR SCA STR ONL TH-I 006 (extr coun may ONL dark grou part neve after corre tions (dou role book ing r flagg D)

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
10	Detection- efficiency mismatch (detector/ basis; time-, wavelength- , mode- dependent)	Single multiplier (TH-PAR- 002) = hidden sub- stitution of the worst case by a mean	Proof- modifying —the canonical detector- side proof gap. Eve’ s mode choice gives knowledge/ control of Bob’ s outcomes without in- tercepting the encoded basis. Attacks: Makarov et al., PRA 74, 022313 (2006); Qi 2007; Zhao 2008; faked states: Makarov & Skaar, QIC 8, 622 (2008); spatial- mode: Sajeed et al., PRA 91, 062301 (2015); turbulence- induced: Chai- wongkhot et al., PRA 99, 062315 (2019) —all VERIFIED	Mismatch- bounded proofs: Fung et al., QIC 9, 131 (2009); Lydersen & Skaar, QIC 10, 60 (2010); Marøy et al., PRA 82, 032337 (2010); Bochkov & Trushechkin, PRA 99, 032308 (2019); Zhang et al., PRR 3, 013076 (2021); Trushechkin, Quantum 6, 771 (2022); Marcomini et al., QST 10, 035002 (2025); basis- dependent detection: Grasselli et al., PR Applied 23, 044011 (2025); satellite- specific: Ivchenko et al., arXiv:2608.09793 (2026) — PREPRINT	Bounded mismatch ratio __min/ __max resolved per mode (time/wave- length/ polariza- tion/ spatial)	Per- detector efficiency maps vs arrival time, , po- larization, spatial mode; inter- detector relative calibration with uncer- tainty; temporal stability and rate dependence of mismatch	CI per map over envelope; nonzero key only for bounded mismatch —the bound is a characteri- zation product, not a default	PRO PRO CAL gated UNI CH REC (D’ s C/E cons TH-I 002 t PAR SCA STR ONL unifo sensi swee

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
11	Wavelength- dependent response	TH-PAR- 002 defined at design wavelength only	Proof- modifying: wavelength is Eve- controllable and converts spectral choice into mismatch (row 10) and decoy leakage (Li et al., PRA 84, 062308, 2011 – VERI- FIED; Jiang et al., PRA 86, 032310, 2012 – VERI- FIED). Out-of- band sensitivity is an unmodeled Eve→receiver channel invisible to QBER	Certified spectral filtering with measured out-of- band rejection entered as device as- sumption; mismatch- aware proof bounded over the full spectral acceptance; spectral- content monitoring as engineering counter- measure (not a proof primitive)	Per- detector spectral response (); filter rejection function; mismatch bound over full acceptance band	Per- detector spectral response curves across full sensitivity range; receiver spectral transmis- sion; out-of- band rejection of all filters; wavelength dependence of basis splitter	CI over spectral envelope incl. Doppler- shifted acceptance band	D lab UNI SEC BU (adv chan entir outs curre mod Res UNI SEC BU Class tion ratio vs si rows (red- C5): a cha ized accep band wave depe enter exac rows –Eve contr mod depe cover bound mism proof gate char: zatio (PRO PRO CAN logic stric

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12	Polarization- dependent response	TH-PAR- 002 = polariza- tion average only	Proof- modifying when the polariza- tion response differs between the two detectors of a basis (direct mismatch within the measure- ment basis) or couples bases (basis- dependent detection probability: Grasselli et al., PR Applied 23, 044011, 2025 – VERI- FIED). Common- mode dependence is engineering- only, but common- modeness is a measured property. In a polarization- encoded downlink, polarization- dependent	Mismatch- bounded proofs (row 10 set); explicit receiver po- larimetry (Mueller/ Jones) feeding polarization- resolved POVM; loss- tolerant analysis if encoding imperfec- tions absorbed at source (Marcomini et al., QST 10, 035002, 2025); squashing with flag structure (Gittsovich et al., PRA 89, 012325, 2014)	Stokes- resolved per detector; receiver Mueller matrix; bounded polarization- resolved mismatch	Polarization- resolved efficiency maps per detector; receiver Mueller po- larimetry; temporal/ thermal drift of po- larization response across a pass	CI per Stokes component over the envelope; worst-case feeds mismatch bound	D la PRO PRO CAL + UNI CHA REC Reso PRO PRO CAN gatec UNM CHA REQ TH-I 002 t PAR SCA STR ONL (pola tion avera

Table 1 – continued

#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
13	Detector memory (general cross-pulse effects)	None —and the fixture’ s IID extraneous- count term (TH-PAR- 006) is explicitly forbidden to carry correla- tions	Proof- modifying at the statistical core: slot-IID detection statistics (random sampling / Serfling) underlie the frozen finite-key analysis; any slot-i dependence on slots i (afterpuls- ing, recovery, charge ac- cumulation, TDC pipeline, electronic crosstalk) breaks it. Detector- side correlated frameworks are thinner than source-side: Nahar & Lütken- haus, arXiv:2503.06328 (2025) — PREPRINT; Tupkary et al., Quantum 9, 1937 (2025) —	Martingale- based finite-key statistics over the actual correlated click process (Azuma 1967; Kato arXiv:2002.04357); explicit receiver state- machine with bounded memory length and worst-case condition- ing; abort criteria on observed correlation statistics	Bounded memory length; worst-case conditional probabili- ties per receiver state	Higher- order click- correlation functions per detector and across detectors (g^2 , lag- resolved conditional probabili- ties); inter- channel crosstalk maps; history- dependence stress tests	Statistical tools verified; complete correlated- detector proof for this fixture: absent (§6)	D la UNI PRO REC (no c agree

Table 1 – continued

#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
14	Characterization — the uncertainty (all proof-relevant parameters)	None — the fixture treats every input as an exact constant	Meta-level proof-modifying: proof parameters are statistical estimates; treating estimates as exact re-introduces the point-value fallacy one level up, understates the composed by $\Sigma _j$, and can run a device outside S_robust as if inside. Framework: Tan & Nahar, PRX Quantum 7, 020342 (2026) — VERIFIED (robust parameter set; per-parameter CIs at $1 - _j$; reject if any interval exits S_robust ; joint-	Certify-then-run architecture (Tan-Nahar 2026): Clopper-Pearson-type intervals per parameter type; reject-and-abort; additive epsilon bookkeeping; partial-characterization numerics: Currás-Lorenzo et al., QST 10, 035031 (2025) — secondary-verified	Robust parameter set S_robust ; per-parameter CIs $[\hat{low}, \hat{upp}]$; failure probabilities $_j$; $_char = \Sigma _j$	Per parameter: measurement protocol, sample size, confidence construction, $_j$; envelope coverage per the repetition matrix (time, temperature, wavelength, polarization, optical power, count rate, device age)	Union bound (Bonferroni) over all parameters $\times$ envelope cells; no independence assumption	UN CH REC (defin tiona and agree zero teriz this rend every secu relev outp cond comp tion the f s em num (e.g. N-02 marg supp secu claim

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#	Device effect	Current scalar model	Security relevance	Candidate proof treatment	Required mathemati- cal parameter	Required characteri- zation evidence	Confidence treatment	Curr statu
15	Aging / cross- instance drift	None — fixture scalars are time- invariant constants; no epoch structure	Proof- modifying at the meta level: a CI valid at charac- terization epoch t is not evidence at operating epoch t without drift/aging bounds; an assumed drift model (e.g. linear/ Lipschitz) is itself a proof condition that must be listed in the model class U_models. On-orbit detector aging is measurable in principle (Lenart et al., Commun. Phys. 8, 118, 2025 — secondary- verified, existence evidence only; no Q-Orbit data)	Multi- epoch characteri- zation with worst-case hull of per-epoch CIs; expanding- interval model with an aging margin that is itself char- acterized; Tan–Nahar robust- domain require- ment that the proof cover the whole operating envelope	Aging envelope bound per parameter (interval growth vs time/ radiation/ thermal cycling); hull- construction rule	Multi- epoch characteri- zation campaigns over the claimed validity window; radiation/ thermal- cycling exposure record; per-epoch CIs with own	Additive per campaign/ cell; union bound; parameter UNCHAR- ACTER- IZED beyond the last charac- terized epoch absent a character- ized aging model	UNI CH REC (E' s fail-c rule; cons No fi scala touch this

2.1 Fixture-scalar touch map (TH-PAR-001...008)

Explicit record of which of the 8 software-contract scalars partially touch which effects (scalar inventory per CFR B-07 baseline block; audit §B):

Scalar	Meaning	Touches effect(s)	Nature of contact
TH-PAR-001	additional system loss (dB)	none of the 15	channel-side engineering scalar; not a device-imperfection representation
TH-PAR-002	detector-efficiency multiplier	6, 7, 8, 10, 11, 12	PARTIAL-SCALAR-STRESS-ONLY in each case: a single photon-number-independent, rate-independent, mode-independent, polarization-averaged, design-wavelength, window-averaged value; cannot express mismatch, rate dependence, or mode/time resolution
TH-PAR-003	repetition rate	none directly	sets the rate <i>context</i> in which effects 6, 7, 9, 13 operate, but encodes no response model
TH-PAR-004	signal intensity	adjacent to 3 (via C S3)	exact-point scalar; PARTIAL-SCALAR-STRESS-ONLY for independent fluctuation stress sweeps; no correlation structure (effect 3 untouched)
TH-PAR-005	weak-decoy intensity	adjacent to 3 (via C S3)	as TH-PAR-004
TH-PAR-006	extraneous-count probability	9 (boundary condition)	legitimate ONLY for IID dark/background counts; must never absorb afterpulse or any other correlation (prohibited hidden substitution)
TH-PAR-007	afterpulse probability	9	equilibrium IID marginal only – PARTIAL-SCALAR-STRESS-ONLY; cannot represent history/rate/basis dependence or Eve actuation

Table 2 – continued

Scalar	Meaning	Touches effect(s)	Nature of contact
TH-PAR-008	intrinsic QBER	4 (partially)	absorbs only the basis-independent, stochastic-symmetric component of state-preparation flaws; basis-dependent SPF content is prohibited from this scalar

Consistency note: the fixture’s own audit (AUD-016-010) records a 16-row imperfection mapping with 7 rows UNMAPPED and 7 without substitution; REG-012 confirms no invented penalty terms. The present 15-row matrix supersedes nothing numerically — it is the proof/characterization-level expansion of that fixture-level register, and its row-level CSV verification remains gated by CFR REQ-05.

3. Interaction flags

Merged from Agent C (§Master-Table interaction flags) and Agent D (rate-dependence cluster). These are joint effects that per-row analysis understates; any V0.17 proof-profile selection must treat them jointly.

1. **S2×S4 (effects 2×3).** Modulator memory typically correlates encoding and intensity jointly; separate per-DOF correlation-length bounds may understate the joint correlation. Treat via joint-correlation analyses: Pereira et al., QST 10, 015001 (2025); Mizutani et al., npj QI 5, 8 (2019).
2. **S5×S4 (effects 1×3).** In gain-switched lasers, intensity correlations co-occur with inter-pulse phase coherence; characterizing one without the other is insufficient — the phase-PDF measurement and the conditional-intensity measurement must be a joint campaign.
3. **S1×S6 (effects 4×5).** A “state-preparation flaw” measured only in the qubit mode can masquerade as a side channel in an unmeasured mode; characterization must specify the measured mode, and the SPF budget and the distinguishability budget must be reconciled against the same mode list.
4. **S7×S3 (effect 5 × TH-PAR-004/005 regime).** Injected Trojan-horse light can shift modulator operating points, *inducing* intensity fluctuations correlated with Eve’s probe — breaking the independence-from-Eve condition that the Mizutani-type fluctuating-intensity treatment (NJP 17, 093011, 2015) requires. Intensity-interval characterization must therefore be performed under injected-light stress, not only in quiet operation.
5. **D rate-dependence cluster (effects 6×7×9×10×13, with TH-PAR-003 as rate context).** Dead time, recovery, saturation, afterpulse trap loading, and effective-efficiency nonlinearity are one coupled response surface driven by instantaneous count rate — which sweeps with the pass loss profile (CFR N-01 baseline window 102 s over a 693-sample pass; loss range spans the frozen 693-point curve). Consequences: (i) effective efficiency is time-varying within a pass, so a single per pass is an equilibrium fiction; (ii) rate-dependent yields threaten the decoy method’s central identity (photon-number yield intensity-independence) — see §6 open problems; (iii) differential detector loading dynamically manufactures basis-dependent mismatch even from initially matched

detectors. The cluster must be characterized as one rate-swept campaign, not five independent ones.

4. Hidden-substitution watchlist

Terms that must **never** be absorbed into scalar QBER / loss / efficiency / IID-noise terms. Each entry is a prohibited move already flagged by C, D, or E; V0.17 software must refuse them (Deliverable 6 §6).

1. **Basis-dependent state-preparation flaws → QBER scalar (TH-PAR-008).** Absorption assumes basis-independent, stochastic-symmetric flaws; the $_\text{X}$ bound is then silently violable. (C §1)
 2. **Correlations → scalar intensity jitter (TH-PAR-004/005).** Widening a jitter scalar manufactures IID randomness where the physics has memory; understates the decoy-estimation failure probability and erases the setting-revealing correlation structure. (C §4; E Category 3)
 3. **Nominal intensity → fluctuation bound.** A nominal $_\text{intensity}$ with no error bar is not a proof input (Tan–Nahar point-value prohibition); exact $\{_\text{j}\}$ make the decoy estimates conditional point computations. (C §3; E §1.1)
 4. **Imperfect phase randomization → added QBER.** PR imperfection changes the channel structure (existence of $_\text{n}$), not the error rate. (C §5; E Category 3)
 5. **Afterpulse correlations → IID extraneous-count term (TH-PAR-006).** Correlations laundered into IID p_ext are invisible at the level of marginals while invalidating the finite-key statistics; also creates the double-role bookkeeping risk with TH-PAR-007. (D §5, §11)
 6. **Efficiency mismatch → single (TH-PAR-002).** Choosing one efficiency (typically the mean) assumes away Eve’s mode choice; privacy is then computed against a weaker-than-physical adversary. The single clearest hidden substitution. (D §6)
 7. **Dead time / recovery / saturation → constant or “noise” QBER.** Erases pass-profile rate dependence, differential detector occupancy, and the adversarial control channel; pileup mislabeled as noise conceals the attack surface. (D §1–§3)
 8. **Security claims with empty leakage/isolation fields.** The margin equation emits positive key while Eve may hold setting information; prohibited. (C §7; E Category 3 guard)
 9. **Point estimates → confidence intervals.** Any scalar without (CI, $_\text{sigma}$, envelope) is a Category-2 fabrication risk; datasheet “typical” values are refused. (E §1.2, Category 2)
 10. **Drift extrapolation → constant parameters.** Applying a CI outside its validity window/envelope without a characterized aging model. (E §1.4; effect 15)
 11. **Variable-length / adaptive key outputs → fixed-length $_\text{s}$ / $_\text{c}$ semantics.** Length chosen from observed pass statistics needs the variable-length proof (Tupkary et al., PRR 6, 023002, 2024 — VERIFIED), not a label reuse. (E Category 3)
 12. **“Certified secure” / conditional-on-approval language.** Only joint bounds $\text{Pr}[\text{approve insecure}] _\text{char} + _\text{protocol}$ are valid; conditional-on-approval claims are a prohibited probability conflation. (E §1.1, R1)
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5. Engineering-model vs security-proof split

An effect is **count-model-only** exactly while all of the following hold: identical across detectors/bases (common-mode), independent of the signal settings and of Eve’s actions, memoryless at slot granularity, and monitored. The moment any condition fails, the effect escalates. Summary:

Effect	Count-model-only regime (non-adversarial)	Escalation trigger → proof-modifying
6 dead time / recovery	Uniform per-detector μ_d , recovery provably complete between slots, honest rates only	Rate variation across pass (dynamic mismatch); any adversarial rate manipulation (Weier 2011); recovery incomplete between slots (memory)
7 saturation	Inside certified linear range, identical across detectors, monitored, honest flux only	Differential saturation; any approach by injected light (blinding-class entry)
8 timing jitter	Perfectly symmetric jitter, mode-matched detectors	Any inter-detector/basis jitter asymmetry (time-shift enabler); jitter leaking into public timing (side channel)
9 afterpulsing	Stationary equilibrium IID marginal only (rate constant)	Any rate variation, history conditioning, basis correlation, or Eve actuation
10 mismatch	None —mismatch is proof-relevant by definition; only its <i>absence</i> (characterized equality) is engineering	Always proof-side; bounded-mismatch proofs need measured bounds
11 wavelength dependence	None at design if out-of-band rejection is characterized and certified	Any uncharacterized out-of-band response
12 polarization dependence	Strictly common-mode response (measured, not assumed)	Detector-differential or basis-coupling response
1–5, 13 (source side, detector memory)	Never count-model-only	Always proof-side: they alter the source/detection model or the finite-key statistics, not merely rates
14, 15 (characterization, aging)	Never count-model-only	Meta-level: govern whether any number is a bound at all

6. Genuine open proof problems (literature gaps, not Q-Orbit failures)

- 1. Detector-side correlated afterpulsing in finite-key decoy proofs.** Verified physical models of history-dependent afterpulsing exist (Ziarkash 2018; Itzler 2012; Horoshko 2017), and martingale concentration tools exist (Azuma 1967; Kato 2020), but no verified literature delivers a complete finite-key decoy-state security proof with correlated afterpulse noise on the detection side for this fixture class. Source-side correlation frameworks (Pereira 2020; Sixto 2022; Pereira QST 2025) do not transfer directly to the detection side. Also open: whether worst-case IID over-approximation of afterpulse correlation is provably pessimistic for key rate (such a theorem must not be assumed — D §5).
- 2. Rate-dependent yields inside the decoy method.** The decoy identity (intensity-independent photon-number yields) fails whenever $\mu_d = (\text{rate})$; a turnkey re-derived decoy bound with rate-

dependent yields was not found in verified literature (D §10). Certified-linear-range operation plus monitoring is the only currently verified fallback.

3. **Full composable integration of certification.** Tan–Nahar (PRX Quantum 7, 020342, 2026) Appendix C notes some technical aspects of full composable integration of certification into Abstract-Cryptography frameworks remain open; Q-Orbit adopts the conservative union bound (Deliverable 6 §4) instead.
4. **Complete correlated-memory detector treatment.** Partial tools exist (Nahar & Lütkenhaus, arXiv:2503.06328, 2025 — PREPRINT; Tupkary et al., Quantum 9, 1937, 2025 — VERIFIED) but no complete bounded-memory detector model inside a finite-key decoy proof for this fixture.

These gaps constrain which proof profiles V0.17 can select for effects 9, 13 and the rate-dependence cluster; they are documented here so that no future phase mistakes them for resolved items or for Q-Orbit-specific defects.

7. Consolidated verdict

- 0 of 15 effects are **MAPPED-IN-CURRENT-FIXTURE**. **5 rows** resolve to **UNMAPPED-PROOF-REQUIRED** (1, 2, 4, 9, 13 — row 4 for its basis-dependent part), **2 rows** are **BLOCKING** (5, 7) pending any characterization, **2 rows** are **UNMAPPED-SECURITY-BUDGET** (6, 11 — both strictest resolutions), and the remaining 6 rows are **PROOF-PROFILE-CANDIDATE** gated by **UNMAPPED-CHARACTERIZATION-REQUIRED** (3, 10, 12) or **UNMAPPED-CHARACTERIZATION-REQUIRED** (8, 14, 15), exactly as tabulated in §2. (Red-team C5 correction: an earlier verdict text read “4 rows ... (1, 2, 9, 13 — with row 4 also ...)” — self-contradictory, the listed set plus row 4 is 5 — and counted only row 6 as USB, silently dropping row 11’s USB label.)
- Every proof-profile candidate is gated by characterization evidence that does not exist (CFR B-01). Under the fail-closed rule, all numerical outputs of the frozen fixture (including CFR N-02/N-03) are conditional computations supporting no security claim.
- All references above are carried from the verified registers of the C/D/E briefs only; preprints (Marwah–Dupuis; Trefilov; Burenkov; Ivchenko; Nahar–Lütkenhaus 2025; arXiv:2603.03217) are labeled PREPRINT and are non-load-bearing.